

CURRICULUM VITÆ

Jeffrey Vincent Backus

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RESEARCH FOCUS

As a high-energy theorist, I am broadly interested in exploring topics in theoretical physics with direct experimental consequences. In this vein, my current work focuses on two closely related themes. First, I study the surprisingly rich combinatorial and geometrical structure of scattering amplitudes in quantum field theory and string theory. I am primarily interested in the application of these novel techniques to realistic models of the universe. Second, I contribute to phenomenological studies of future collider experiments, especially to that of a proposed multi-TeV muon collider.

EDUCATION

Princeton University || Graduate School

Doctor of Philosophy, Physics

Princeton, NJ

2023 -

- Centennial Fellow and NSF Graduate Research Fellow
- Thesis Advisor: Nima Arkani-Hamed

Cornell University || College of Arts and Sciences

Bachelor of Arts, Physics and Mathematics, Summa Cum Laude

Ithaca, NY

2019 - 2023

- Honors thesis: *Towards an optimal extraction of γ from $B \rightarrow DK$ decay modes*
- Bachelor Advisor: Yuval Grossman
- Cumulative GPA: 4.160 / 4.000

PUBLICATIONS

Full list may also be found on [INSPIRE-HEP](#)

1. J.V. Backus

New Recursions for the Canonical Scalar-Scaffolded Yang-Mills Amplitude

[arXiv:2510.19901](#) [hep-th]

2. J.V. Backus and C. Figueiredo

Surface gauge invariance, soft factors and the transmutation of gluons into scalars

JHEP **09** (2025) 069

[arXiv:2505.17179](#) [hep-th]

3. J.V. Backus and L. Rodina

Emergence of Unitarity and Locality from Hidden Zeros at One-Loop Order

Phys. Rev. Lett. **135** (2025) 13, 131601

[arXiv:2503.03805](#) [hep-th]

4. J.V. Backus, M. Freytsis, Y. Grossman, S. Schacht, and J. Zupan

Toward extracting γ from $B \rightarrow DK$ without binning

Eur. Phys. J. C **83** (2023) 877

[arXiv:2211.05133](#) [hep-ph]

INVITED TALKS

1. *Soft Limits and the Transmutation of Gluons to Scalars*

April 2025

IAS Amplitudes Seminar	Institute for Advanced Study, Princeton, NJ
2. <i>What is Kinematic Space?</i> Amplitudes and Combinatorics Synergy Event	October 2024 Institute for Advanced Study, Princeton, NJ
3. <i>Scalar-Scaffolded Gluons and Soft Factors</i> UNIVERSE+ Kick-Off Meeting	August 2024 Physikzentrum, Bad Honnef, Germany
4. <i>Using Machine Learning Techniques to Measure CP Violation</i> Cornell HEP Experiment-Theory Workshop	February 2023 Cornell University, Ithaca, NY
5. <i>Measuring CP Violation with $B \rightarrow DK$</i> Cornell-Syracuse HEP Theory Conference	November 2022 Syracuse University, Syracuse, NY

HONORS, AWARDS & FELLOWSHIPS

– NSF Graduate Research Fellowship Program National Science Foundation	2024 - 2029
– U.S. Delegate to the 73rd Lindau Nobel Laureate Meeting Lindau Council	2024
– Undergraduate Teaching Assistant Award Cornell Department of Physics	2023
– College of Arts & Sciences Exceptional Senior Cornell University	2023
– Philip Merrill Presidential Scholarship Cornell University	2023
– Centennial Fellowship Princeton University	2023 - 2028
– Stanford Graduate Fellowship Stanford University (Declined)	2023
– Barry M. Goldwater Scholarship Goldwater Foundation	2022 - 2023
– Phi Beta Kappa Honor Society Cornell University	2022
– Rawlings Cornell Presidential Research Scholarship Cornell University	2021 - 2023
– Research Experience for Undergraduates (REU) Fellowship Stony Brook University	2021

PROFESSIONAL SERVICE

– Member, Diversity and Inclusivity Team ERC UNIVERSE+ Collaboration	2025 -
– Member, Research Data Management Team ERC UNIVERSE+ Collaboration	2024 -
– Co-Organizer, High-Energy Theory Journal Club Princeton Department of Physics	2024 -

TEACHING & ADVISING EXPERIENCE

– Undergraduate Teaching Assistant Cornell Department of Physics	2020 - 2023
– PHYS 4444: Introduction to Particle Physics	
– PHYS 3317: Applications of Quantum Mechanics	
– PHYS 3316: Basics of Quantum Mechanics	
– PHYS 1116: Honors Mechanics and Special Relativity	
– Zoom a Princeton Physicist Mentor Princeton Department of Physics	2024 -
– Undergraduate Physics Major Mentor Princeton Department of Physics	2023 -
– First-Year Physics Major Mentor Cornell Society for Physics Students	2021 - 2023